



8086 Pins & Signals

Chapter 2

8086 Pinout

AD15-AD0: Multiplexed address(ALE=1)/data bus(ALE=0).

A19/S6-A16/S3 (multiplexed): High order 4 bits of the 20-bit address OR status bits S6-S3.

M/IO': Indicates if address is a Memory or IO address.

RD': When 0, data bus is driven by memory or an I/O device.

WR: Microprocessor is driving data bus to memory or an I/O device. When 0, data bus contains valid data.

ALE (Address latch enable): When 1, address data bus contains a memory or I/O address.

GND	---	1	8086	40	---	V_{cc}
AD14	<->	2		39	<->	AD15
AD13	<->	3		38	-->	A16/S3
AD12	<->	4		37	-->	A17/S4
AD11	<->	5		36	-->	A18/S5
AD10	<->	6		35	-->	A19/S6
AD9	<->	7		34	-->	!BHE/S7
AD8	<->	8		33	<--	MN/!MX
AD7	<->	9		32	-->	RD
AD6	<->	10		31	<->	!RQ/!GT0,HOLD
AD5	<->	11		30	<->	!RQ/!GT1,HOLD
AD4	<->	12		29	-->	!LOCK,!WR
AD3	<->	13		28	-->	!S2,M/!IO
AD2	<->	14		27	-->	!S1,DT/!R
AD1	<->	15		26	-->	!S0,!DEN
AD0	<->	16		25	-->	QS0,ALE
NMI	-->	17		24	-->	QS1,!INTA
INTR	-->	18		23	<--	!TEST
CLK	-->	19		22	<--	READY
GND	---	20		21	<--	RESET

8086 Pinout

DT/R' (Data Transmit/Receive): Data bus is transmitting/receiving data.

DEN (Data bus Enable): Activates external data bus buffers.

S7: Logic 1

S6: Logic 0.

S5: Indicates condition of IF flag bits.

GND	---	1	8086	40	---	V_{cc}
AD14	<->	2		39	<->	AD15
AD13	<->	3		38	-->	A16/S3
AD12	<->	4		37	-->	A17/S4
AD11	<->	5		36	-->	A18/S5
AD10	<->	6		35	-->	A19/S6
AD9	<->	7		34	-->	!BHE/S7
AD8	<->	8		33	<--	MN/!MX
AD7	<->	9		32	-->	RD
AD6	<->	10		31	<->	!RQ/!GT0,HOLD
AD5	<->	11		30	<->	!RQ/!GT1,HOLD A
AD4	<->	12		29	-->	!LOCK,!WR
AD3	<->	13		28	-->	!S2,M/!IO
AD2	<->	14		27	-->	!S1,DT/!R
AD1	<->	15		26	-->	!S0,!DEN
AD0	<->	16		25	-->	QS0,ALE
NMI	-->	17		24	-->	QS1,!INTA
INTR	-->	18		23	<--	!TEST
CLK	-->	19		22	<--	READY
GND	---	20		21	<--	RESET

8086 Pinout

S4-S3: Indicate which segment is accessed during current bus cycle:

S4	S3	Function
0	0	Extra segment
0	1	Stack segment
1	0	Code or no segment
1	1	Data segment

S2', S1', S0' : Indicate function of current bus cycle (decoded by 8288)

$\overline{S2}$	$\overline{S1}$	$\overline{S0}$	Function
0	0	0	Interrupt Ack
0	0	1	I/O Read
0	1	0	I/O Write
0	1	1	Halt
1	0	0	Opcode Fetch
1	0	1	Memory Read
1	1	0	Memory Write
1	1	1	Passive

GND	---	1	8086	40	---	V_{cc}
AD14	<->	2		39	<->	AD15
AD13	<->	3		38	-->	A16/S3
AD12	<->	4		37	-->	A17/S4
AD11	<->	5		36	-->	A18/S5
AD10	<->	6		35	-->	A19/S6
AD9	<->	7		34	-->	!BHE/S7
AD8	<->	8		33	<--	MN/!MX
AD7	<->	9		32	-->	RD
AD6	<->	10		31	<->	!RQ/!GT0,HOLD
AD5	<->	11		30	<->	!RQ/!GT1,HOLD A
AD4	<->	12		29	-->	!LOCK,!WR
AD3	<->	13		28	-->	!S2,M/!IO
AD2	<->	14		27	-->	!S1,DT/!R
AD1	<->	15		26	-->	!S0,!DEN
AD0	<->	16		25	-->	QS0,ALE
NMI	-->	17		24	-->	QS1,!INTA
INTR	-->	18		23	<--	!TEST
CLK	-->	19		22	<--	READY
GND	---	20		21	<--	RESET

8086 Pinout

INTR: When 1 and IF=1, microprocessor prepares to service interrupt. INTA becomes active after current instruction completes.

INTA': Interrupt Acknowledge generated by the microprocessor in response to INTR. Causes the interrupt vector to be put onto the data bus.

NMI: Non-maskable interrupt. Similar to INTR except IF flag bit is not consulted and interrupt is vector 2.

CLK: Clock input must have a duty cycle of 33% (high for 1/3 and low for 2/3s)

GND	---	1	8086	40	---	V_{cc}
AD14	<->	2		39	<->	AD15
AD13	<->	3		38	-->	A16/S3
AD12	<->	4		37	-->	A17/S4
AD11	<->	5		36	-->	A18/S5
AD10	<->	6		35	-->	A19/S6
AD9	<->	7		34	-->	!BHE/S7
AD8	<->	8		33	<--	MN/!MX
AD7	<->	9		32	-->	RD
AD6	<->	10		31	<->	!RQ/!GT0,HOLD
AD5	<->	11		30	<->	!RQ/!GT1,HOLD
AD4	<->	12		29	-->	!LOCK,!WR
AD3	<->	13		28	-->	!S2,M/!IO
AD2	<->	14		27	-->	!S1,DT/!R
AD1	<->	15		26	-->	!S0,!DEN
AD0	<->	16		25	-->	QS0,ALE
NMI	-->	17		24	-->	QS1,!INTA
INTR	-->	18		23	<--	!TEST
CLK	-->	19		22	<--	READY
GND	---	20		21	<--	RESET

8086 Pinout

MN/MX': Select minimum (5V) or maximum mode (0V) of operation.

BHE': Bus High Enable. Enables the most significant data bus bits (D15 -D8) during a read or write operation.

READY: Used to insert wait states (controlled by memory and IO for reads/writes) into the microprocessor.

RESET: Microprocessor resets if this pin is held high for 4 clock periods.

Instruction execution begins at FFFF0H and IF flag is cleared.

TEST': An input that is tested by the WAIT instruction. Commonly connected to the 8087 coprocessor

GND	---	1	8086	40	---	V_{CC}
AD14	<->	2		39	<->	AD15
AD13	<->	3		38	-->	A16/S3
AD12	<->	4		37	-->	A17/S4
AD11	<->	5		36	-->	A18/S5
AD10	<->	6		35	-->	A19/S6
AD9	<->	7		34	-->	!BHE/S7
AD8	<->	8		33	<--	MN/!MX
AD7	<->	9		32	-->	RD
AD6	<->	10		31	<->	!RQ/!GT0,HOLD
AD5	<->	11		30	<->	!RQ/!GT1,HOLD
AD4	<->	12		29	-->	!LOCK,!WR
AD3	<->	13		28	-->	!S2,M/!IO
AD2	<->	14		27	-->	!S1,DT/!R
AD1	<->	15		26	-->	!S0,!DEN
AD0	<->	16		25	-->	QS0,ALE
NMI	-->	17		24	-->	QS1,!INTA
INTR	-->	18		23	<--	!TEST
CLK	-->	19		22	<--	READY
GND	---	20		21	<--	RESET

8086 Pinout

HOLD: Requests a direct memory access (DMA). When 1, microprocessor stops and places address, data and control bus in high-impedance state.

HLDA (Hold Acknowledge): Indicates that the microprocessor has entered the hold state.

RO'/GT1' & RO'/GT0': Request/grant pins request/grant direct memory accesses (DMA) during maximum mode operation.

LOCK': Lock output is used to lock peripherals off the system. Activated by using the LOCK: prefix on any instruction.

GND	---	1	8086	40	---	V_{cc}
AD14	<->	2		39	<->	AD15
AD13	<->	3		38	-->	A16/S3
AD12	<->	4		37	-->	A17/S4
AD11	<->	5		36	-->	A18/S5
AD10	<->	6		35	-->	A19/S6
AD9	<->	7		34	-->	!BHE/S7
AD8	<->	8		33	<--	MN/!MX
AD7	<->	9		32	-->	RD
AD6	<->	10		31	<->	!RQ/!GT0,HOLD
AD5	<->	11		30	<->	!RQ/!GT1,HOLA
AD4	<->	12		29	-->	!LOCK,!WR
AD3	<->	13		28	-->	!S2,M/!IO
AD2	<->	14		27	-->	!S1,DT/!R
AD1	<->	15		26	-->	!S0,!DEN
AD0	<->	16		25	-->	QS0,ALE
NMI	-->	17		24	-->	QS1,!INTA
INTR	-->	18		23	<--	!TEST
CLK	-->	19		22	<--	READY
GND	---	20		21	<--	RESET

8086 Pinout

QS1 & QS0: The queue status bits show status of internal instruction queue. Provided for access by the numeric coprocessor (8087).

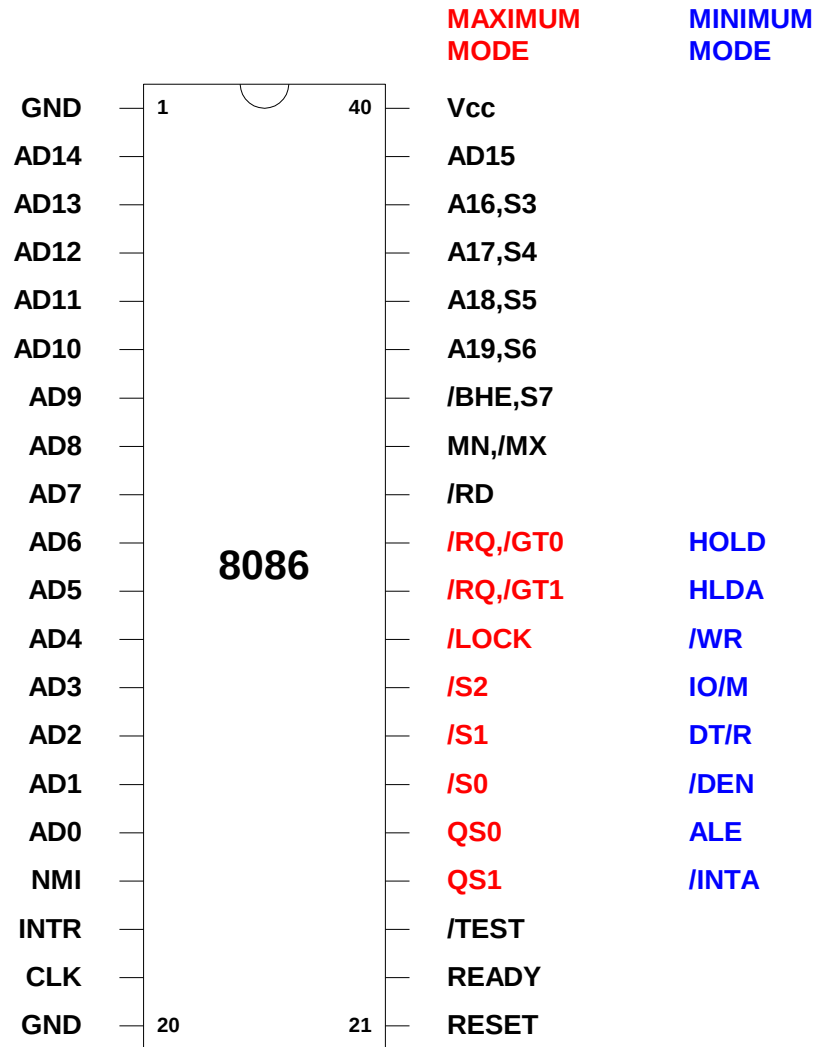
QS1	QS0	Function
0	0	Queue is idle
0	1	First byte of opcode
1	0	Queue is empty
1	1	Subsequent byte of opcode

GND	---	1	8086	40	---	V_{cc}
AD14	<->	2		39	<->	AD15
AD13	<->	3		38	-->	A16/S3
AD12	<->	4		37	-->	A17/S4
AD11	<->	5		36	-->	A18/S5
AD10	<->	6		35	-->	A19/S6
AD9	<->	7		34	-->	!BHE/S7
AD8	<->	8		33	<--	MN/!MX
AD7	<->	9		32	-->	RD
AD6	<->	10		31	<->	!RQ/!GT0,HOLD
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AD4	<->	12		29	-->	!LOCK,!WR
AD3	<->	13		28	-->	!S2,M/!IO
AD2	<->	14		27	-->	!S1,DT/!R
AD1	<->	15		26	-->	!S0,!DEN
AD0	<->	16		25	-->	QS0,ALE
NMI	-->	17		24	-->	QS1,!INTA
INTR	-->	18		23	<--	!TEST
CLK	-->	19		22	<--	READY
GND	---	20		21	<--	RESET

RESET# Signal

- The Active low RESET# signal puts the 8086 into a defined state
 - Clears the flags register, segment registers etc.
 - Sets the effective program address to 0FFFF0h (CS=0F000h, IP=0FFF0h)
- Microprocessor resets if this pin is held high for 4 clock periods.

8086 Microprocessor



Minimum Mode:

- 8086 is configured to support small, single processor systems using a few devices that use the system bus

Maximum Mode:

- 8086 is configured to support multi-processor systems.
- 8086 needs *at least* the following:
 - 8288 Bus Controller,
 - 8284A Clock Generator,
 - 74HC373s and 74HC245s

DMA with 8086 in Minimum Mode

Two signals are used to request/ack a DMA transfer:

- HOLD is an input to the 8086 that requests a DMA action.
- HLDA' is an output from the 8086 granting the DMA action.

The microprocessor responds by suspending the execution of the program and by placing its address, data and control bus in high-impedance states.

8086 Maximum Mode

For systems with more than one processor (multiprocessor systems)

- Processors can execute concurrently
 - Increased performance
- Can have special-purpose processors (e.g., numeric coprocessor, I/O processor)

Some issues:

- Resource sharing
- Synchronization
- Bus contention

But processors synchronize activities

Communicate with each other (IPC)

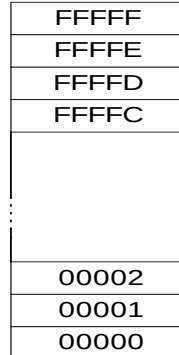
Share resources with each other

Avoid interference (bus contention) with each other, so there must be more control signals in Max mode

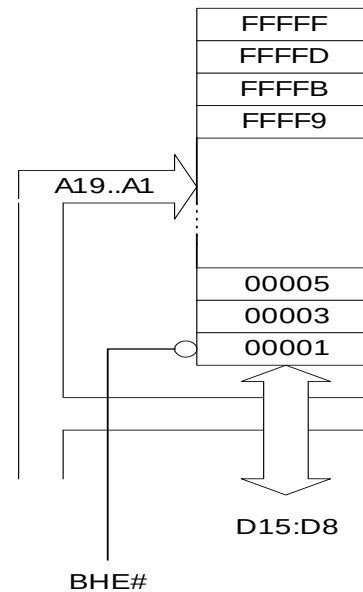
Use of BHE#/A0(BLE#)

BHE#	A0/BLE#	Selection
0	0	Whole word (16-bits)
0	1	High byte to/from odd address
1	0	Low byte to/from even address
1	1	No selection

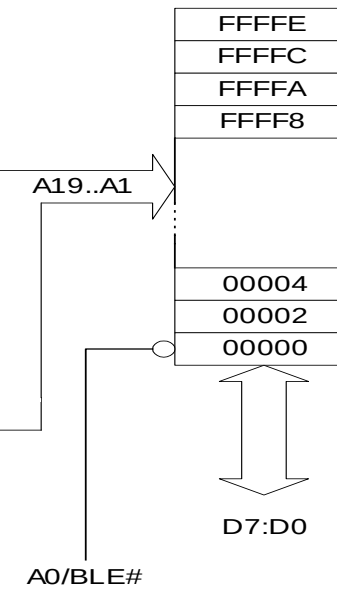
Byte-Wide addressing
(8088)



ODD Addresses (8086)



EVEN Addresses (8086)

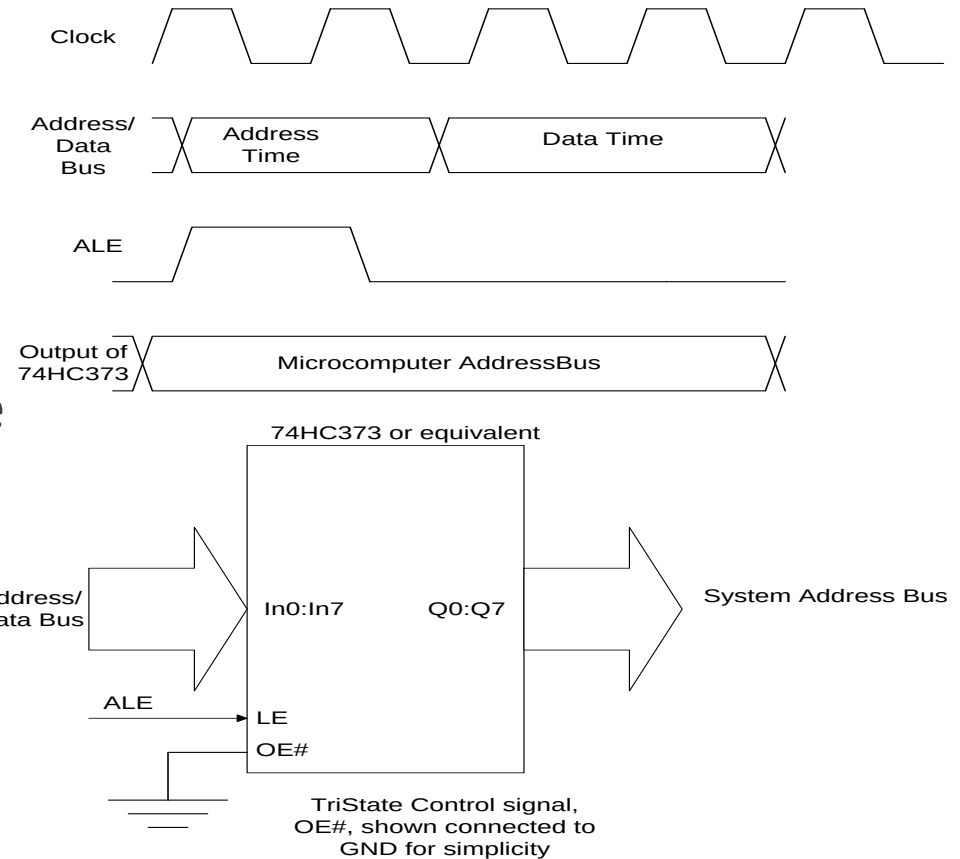


ALE and Address/data Bus Multiplexing

8086 Multiplexes the Address and Data signals onto the same set of pins

Need off-chip logic to separate the signals

Transparent latches designed just for address demultiplexing



Bus Cycle

Timing:

- Each BUS CYCLE on the 8086 equals four system clocking periods (T states).
- The clock rate is 5MHz , therefore one Bus Cycle is 800ns .
- The transfer rate is 1.25MHz .

Memory specs (memory access time) must match constraints of system timing.

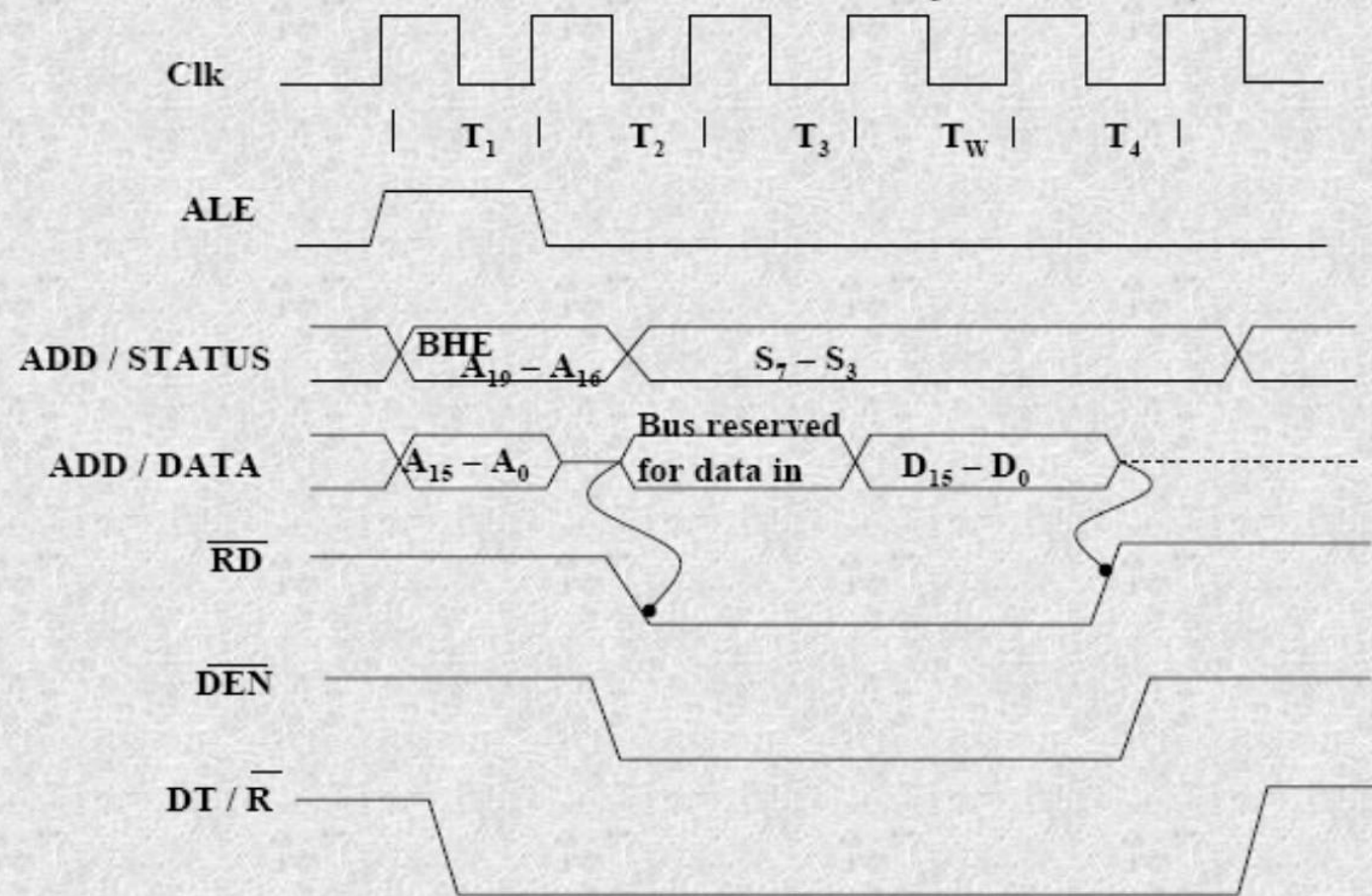
For example,

- bus timing for a read operation shows almost 600ns are needed to read data.
- However, memory must access faster due to setup times, e.g. Address setup and data setup.
- This subtracts off about 150ns .
- Therefore, memory must access in at least 450ns minus another 30-40ns guard band for buffers and decoders

Minimum Mode Pins

$\text{IO}/\overline{\text{M}}$ or $\overline{\text{M}}/\text{IO}$	The $\text{IO}/\overline{\text{M}}$ (8088) or the $\overline{\text{M}}/\text{IO}$ (8086) pin selects memory or I/O.
$\overline{\text{WR}}$	The write line is a strobe that indicates that the 8086/8088 is outputting data to a memory or I/O device.
$\overline{\text{INTA}}$	The interrupt acknowledge signal is a response to the INTR input pin.
ALE	Address latch enable shows that the 8086/8088 address/data bus contains address information.
$\text{DT}/\overline{\text{R}}$	The data transmit/receive signal shows that the microprocessor data bus is transmitting ($\text{DT}/\overline{\text{R}} = 1$) or receiving ($\text{DT}/\overline{\text{R}} = 0$) data.
DEN	Data bus enable activates external data bus buffers.
HOLD	The hold input requests a direct memory access (DMA).
$\overline{\text{HLDA}}$	Hold acknowledge indicates that the 8086/8088 has entered the hold state.
$\overline{\text{SS0}}$	The $\overline{\text{SS0}}$ status line is equivalent to the S0 pin in maximum mode operation of the microprocessor. This signal is combined with $\text{IO}/\overline{\text{M}}$ and $\text{DT}/\overline{\text{R}}$ to decode the function of the current bus cycle .

Minimum Mode 8086 System (cont..)



Read Cycle Timing Diagram for Minimum Mode

Minimum mode - READ

□ Hence the timing diagram can be categorized in two parts,

- the timing diagram for read cycle
- the timing diagram for write cycle.

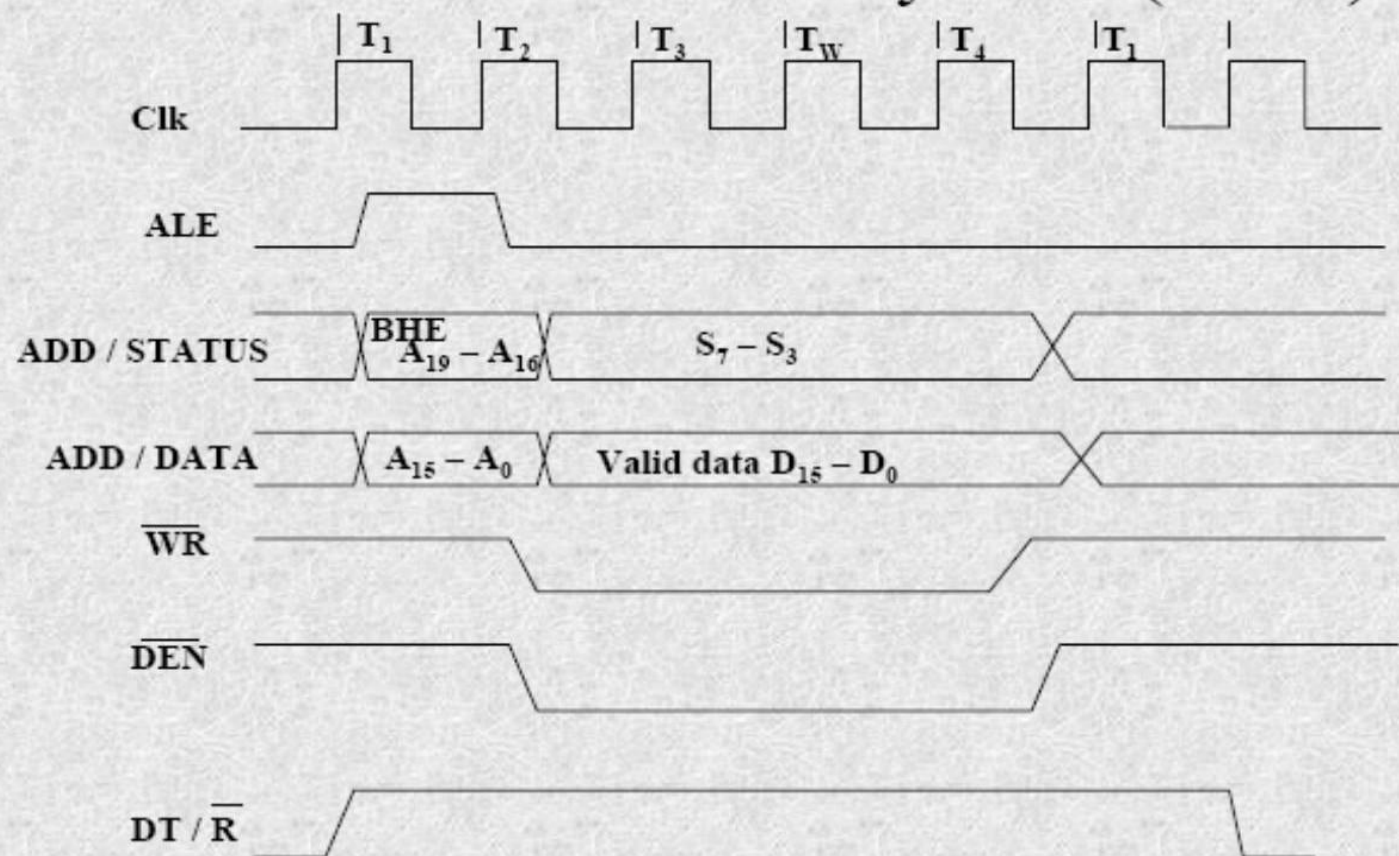
□ The **read cycle** begins in T1 with the assertion of address latch enable (ALE) signal and also M / IO signal.

□ During the negative going edge of this signal, the valid address is latched on the local bus.

Minimum mode – READ continue...

- The BHE and A0 signals address low, high or both bytes. From T1 to T4, the M/IO signal indicates a memory or I/O operation.
- At T2, the address is removed from the local bus and is sent to the output. The bus is then tri-stated. The read (RD) control signal is also activated in T2.
- The read (RD) signal causes the address device to enable its data bus drivers. After RD goes low, the valid data is available on the data bus.
- The addressed device will drive the READY line high. When the processor returns the read signal to high level, the addressed device will again tri-state its bus drivers.

Minimum Mode 8086 System (cont..)



Write Cycle Timing Diagram for Minimum Mode

Minimum mode -Write

- A write cycle also begins with the assertion of ALE and the emission of the address. The M/IO signal is again asserted to indicate a memory or I/O operation. In T2, after sending the address in T1, the processor sends the data to be written to the addressed location.
- The data remains on the bus until middle of T4 state. The WR becomes active at the beginning of T2 (unlike RD is somewhat delayed in T2 to provide time for floating).
- The BHE and A0 signals are used to select the proper byte or bytes of memory or I/O word to be read or write.
- The M/IO, RD and WR signals indicate the type of data transfer as specified in table below.

Maximum mode 8086 system continue...

- In the maximum mode, the 8086 is operated by strapping the MN/MX pin to ground.
- In this mode, the processor derives the status signal S2, S1, S0. Another chip called bus controller derives the control signal using this status information .
- In the maximum mode, there may be more than one microprocessor in the system configuration.
- The components in the system are same as in the minimum mode system.

Maximum-Mode Interface (cont.)

□ 8288 bus controller

S2	S1	S0	<i>Function</i>
0	0	0	Interrupt acknowledge
0	0	1	I/O read
0	1	0	I/O write
0	1	1	Halt
1	0	0	Opcode fetch
1	0	1	Memory read
1	1	0	Memory write
1	1	1	Passive

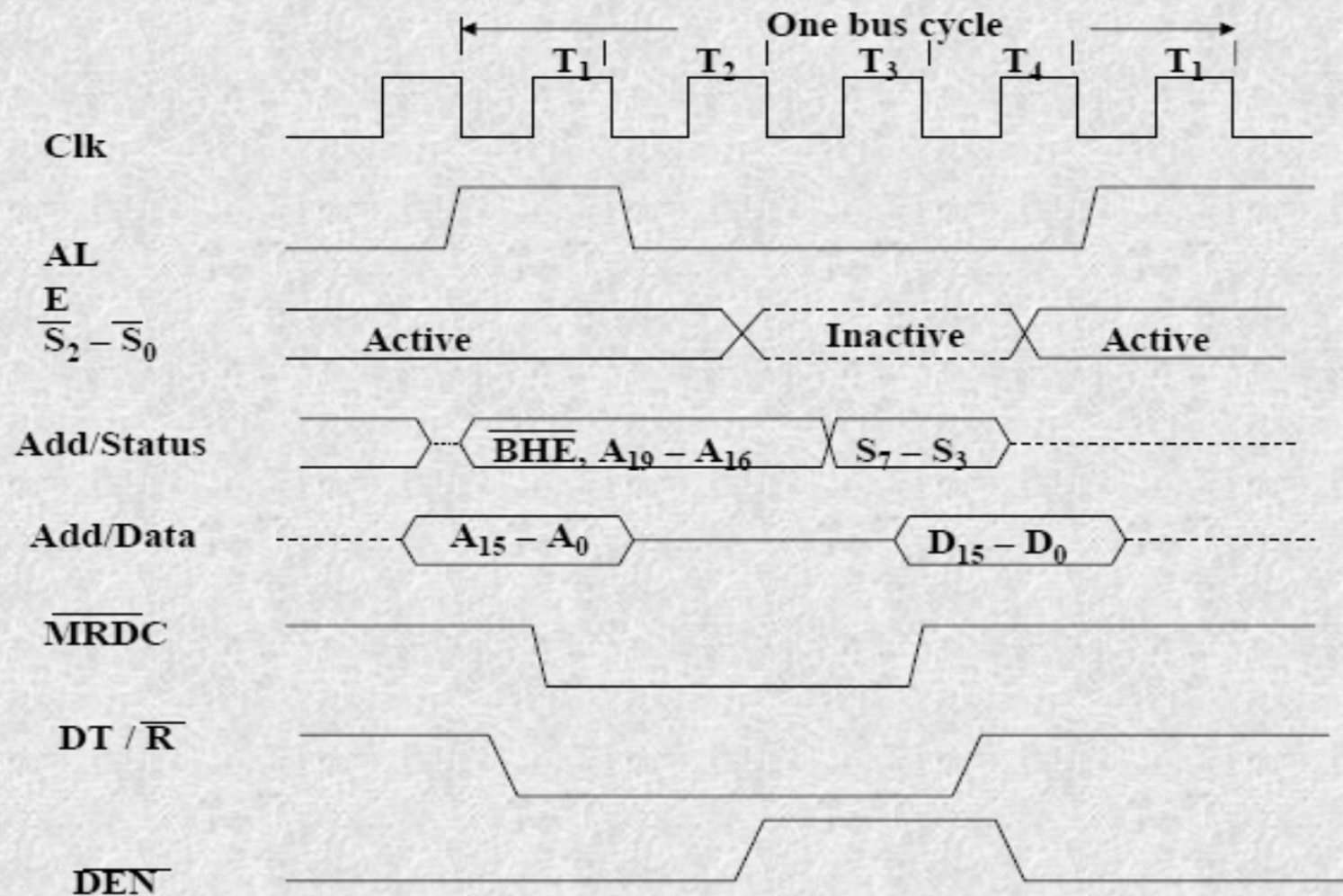
Bus status code

- Maximum-mode memory control signals
 - **MRDC – Memory Read Command**
 - **MWTC – Memory Write Command**

- **AMWC – Advanced Memory Write**

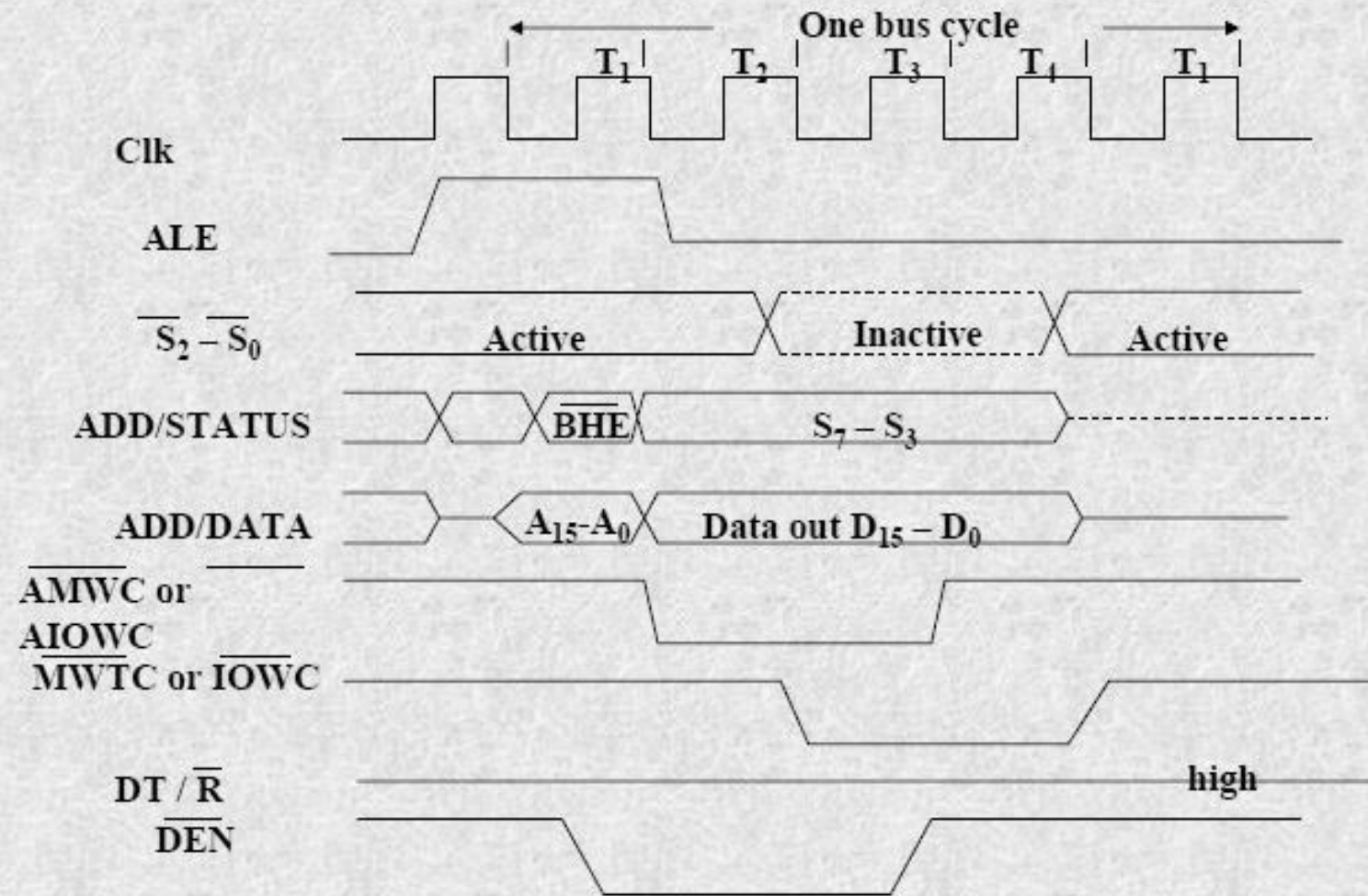
Status Inputs			CPU Cycle	8288 Command
$\overline{S}_2$	$\overline{S}_1$	$\overline{S}_0$		
0	0	0	Interrupt acknowledge	$\overline{INTA}$
0	0	1	Read I/O port	$\overline{IORC}$
0	1	0	Write I/O port	$\overline{IOWC}, \overline{AIOWC}$
0	1	1	Halt	None
1	0	0	Instruction fetch	$\overline{MRDC}$
1	0	1	Read memory	$\overline{MRDC}$
1	1	0	Write memory	$\overline{MWTC}, \overline{AMWC}$
1	1	1	Passive	None

Maximum Mode 8086 System (cont..)



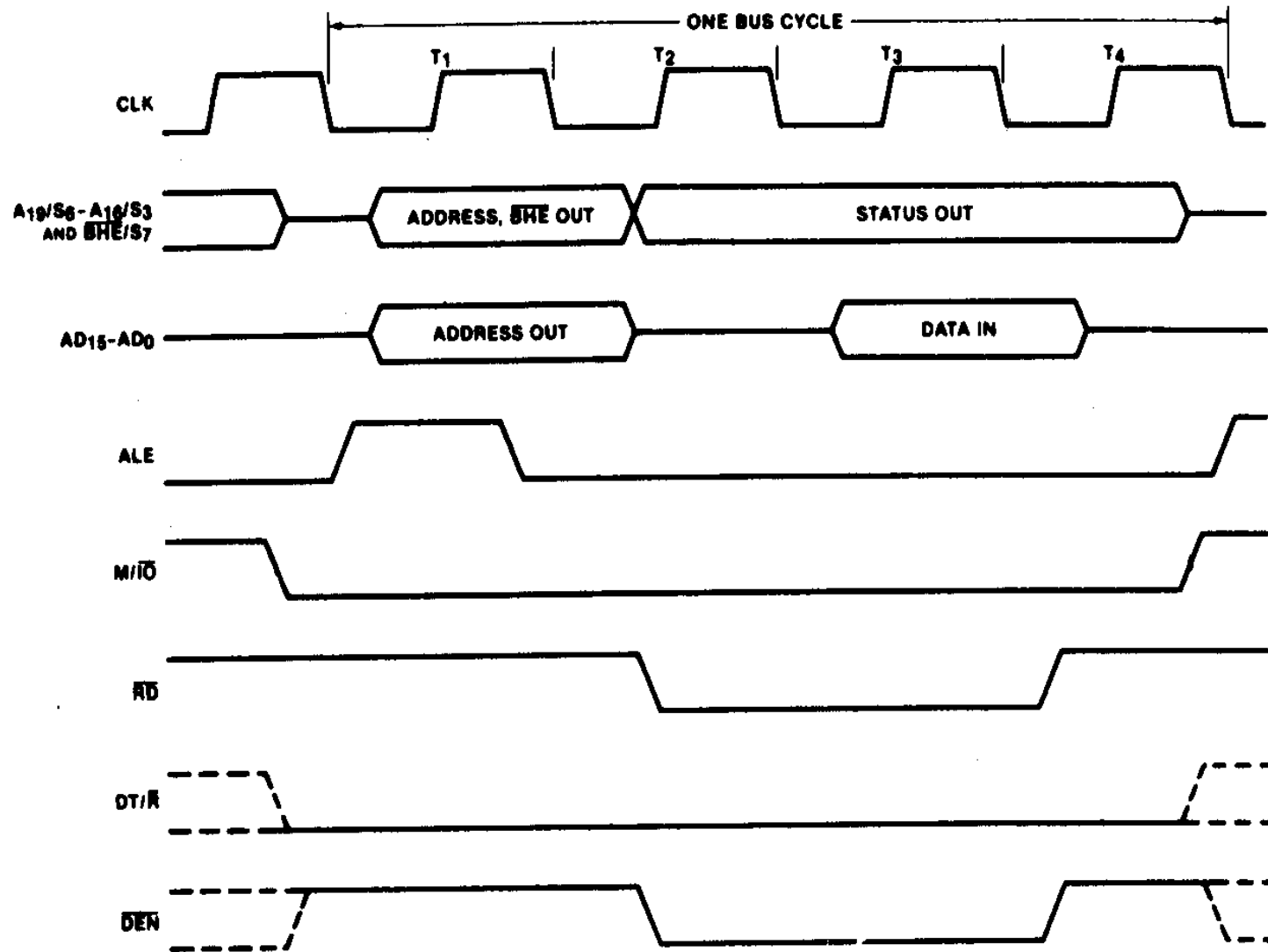
Memory Read Timing in Maximum Mode

Maximum Mode 8086 System (cont..)

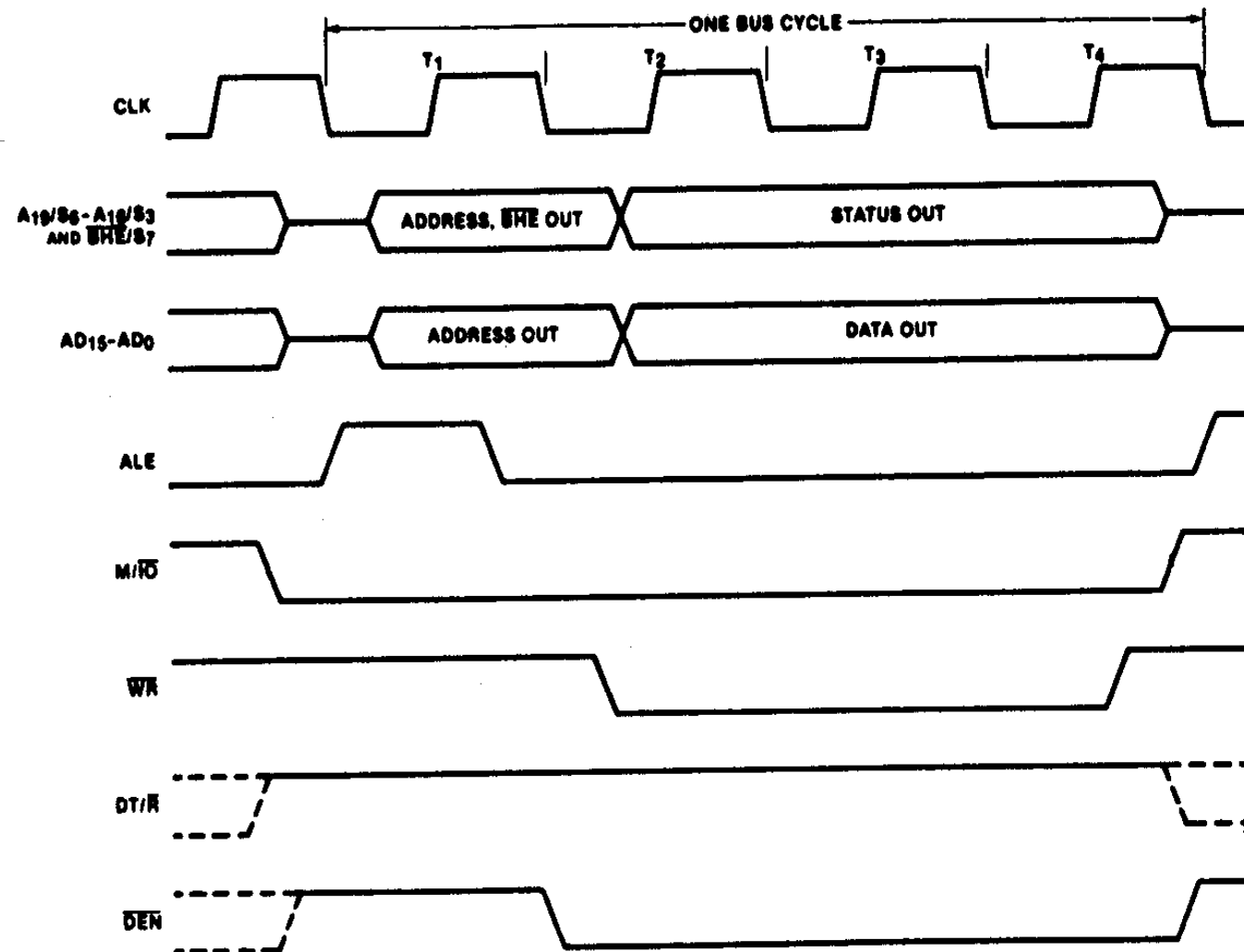


Memory Write Timing in Maximum mode.

Input/Output Bus Cycles (cont.)



Input/Output Bus Cycles (cont.)



Thank You

A solid teal-colored horizontal bar at the bottom of the slide, with a slight gradient from left to right.